

# MCSD Project

Concept Paper

Akos Eröss + 84655 + Module: LCD Module (HD44780-compatible)

Daniel Lidy + 12123429 + Module: Range Sensor (VL53LO/1XV2):

## Used Modules:

List and briefly describe the functionality of the used modules. Please also list the official name of the Clickboard. You can find this in the Click-Module table within the course.

Module 1 :

**16x2 Character LCD Module (HD44780-compatible)** equipped with a pre-soldered **I<sup>2</sup>C Backpack Interface (Funduino / PCF8574)**.

### Key Specifications

- **Display Capacity:** 16 characters per row across 2 lines (alphanumerics and basic custom symbols).
- **Driver Standard:** Industry-standard HD44780 command set.
- **Interface:** I<sup>2</sup>C bus (converts the traditional 16-pin parallel interface down to just 4 wires).

### Backpack Hardware Elements

- **4-Pin Header:** **GND**, **VCC** (typically 5V), **SDA**, and **SCL** for direct microcontroller hookup.
- **Blue Potentiometer:** Onboard trimmer to adjust display text contrast.
- **Jumper Cap (Green/Black):** Enables or disables the LCD backlight (can be removed to control the backlight via an external pin).

Module 2 :

**Time-of-Flight (ToF) Laser Distance / Ranging Sensor** breakout board (specifically for the STMicroelectronics **VL53L0X** or **VL53L1X** module).

### Key Features

- **Operating Principle:** Measures distance by timing how long invisible 940 nm infrared laser light takes to bounce back from an object, providing millimetre-level accuracy regardless of target color or reflectivity.
- **Range:** Up to ~2 meters (VL53L0X).
- **Interface:** I<sup>2</sup>C bus.

### Pinout Breakdown

- **VIN:** Power input (typically 3.3V to 5V).
- **GND:** Ground.
- **SCL / SDA:** I<sup>2</sup>C Clock and Data lines for communication.
- **GPIO1:** Programmable interrupt output (can alert your microcontroller when a threshold distance is crossed).
- **XSHUT:** Active-low shutdown pin (used for power-saving or for readdressing when running multiple sensors on a single I<sup>2</sup>C bus).

## Main Function Description and Block Diagram of the System

Please describe the planned functionality in this section. Detail what capabilities or demonstrations will be possible upon the project's completion. Identify and elaborate on at least three distinct scenarios, each of which must be demonstrated during the presentation. **Additionally, include a block diagram** of the system to illustrate its structure. In your diagram, specify the interfaces involved and explain how data is transferred or communicated between the different modules.

This happens after switching on:

When I turn on the device, the modules are initialized. The output module shows ... The microcontroller initializes the distance sensor, LCD display, buttons, potentiometer and other required peripherals. The LCD shows "System Ready" to indicate that the device is ready to perform measurements.

Scenario 1:

When I operate switch A, then ... when the user presses SW1, the LCD is turned on. When pressing the button second time the system starts a distance measurement. The distance sensor sends the measured value to the microcontroller, which is displayed by the screen. Pressing the button a third time results in the current value freezing as long as the user presses the button again, while the LCD is on.

Scenario 2:

When the value of the sensor changes like this, then ... When the measured distance changes during a measurement, the LCD continuously refreshes the value of the current distance in centimeters.

Scenario 3:

When I don't press any button for 30 seconds,... the device goes into sleep mode the range sensor stops measuring and the LCD turns off. When waking up the screen by pressing SW1 button and the system went into sleep mode after a value has been frozen, the same value should be displayed. Additionally a countdown is displayed on the LCD to inform the user about the on time.

### Nucleo32 Click-Shield (FHTW 2025 v3)

SW1 Push Button  
(Digital In)

GPIO (PA12 / D2)

STM32L432KC MCU

### Slot 2 Header (UART / I2C)

5V

I2C1\_SCL

I2C1\_SDA

3V3

SCL  
/PB6

SDA  
(PB7)

GND

### VL53L0X / VL53L1X Module

GND

ToF Sensor IC  
(Addr: 0x29)

VIN

SCL

SDA

3.3V

SCL

SDA

GND

5.0V

SCL

SDA

GND

### 16x2 Character LCD Unit

#### Funduino I2C Backpack

VCC

SCL

SDA

GND

PCF8574 IO Expander  
(Addr: 0x27 / 0x3F)

4-bit Parallel Bus  
(RS, E, D4-D7, LED)

16x2 HD44780  
Character LCD Panel

## Optional Extended Function Description

The user can switch between different menus using the 2x SW1 -> **potentiometer**.

- **Menu 1 – Distance Measurement:**

The current distance is displayed on the LCD. By pressing **SW1**, the user can save the current measured distance.

- **Menu 2 – Measurement History:**

The LCD displays the **last three saved measurements**.

- **Menu 3 – Sorted Measurements:**

The saved measurement values are **sorted and displayed in ascending order**.